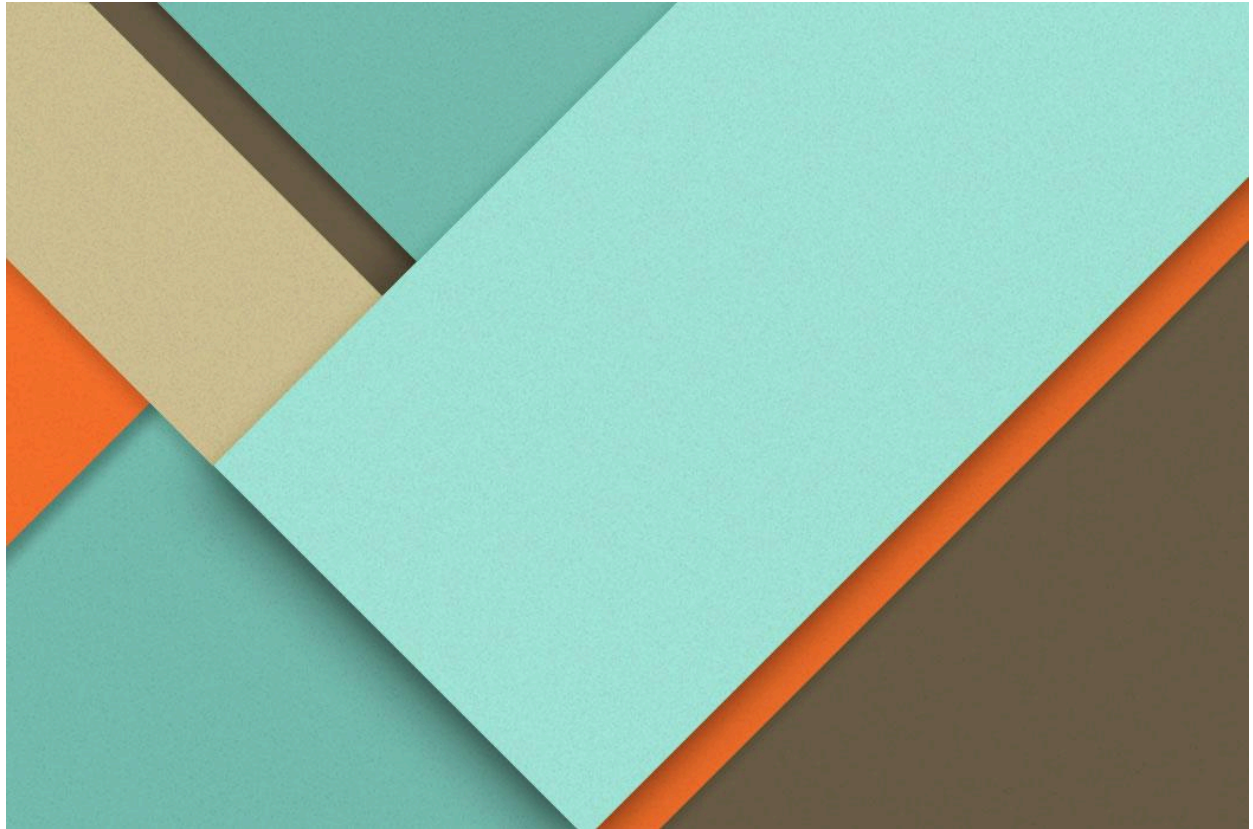




Self-Hosted Matrix Messaging Infrastructure with AI Context Integration

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Overview

We currently use Element, a chat client built on the Matrix protocol, with accounts hosted on the public matrix.org server. While this setup enables decentralized communication, it limits control over data and restricts deeper integration with internal systems.

This project proposes exploring the deployment of a self-hosted Matrix server (Synapse) to manage messaging infrastructure internally. Also we see how to integrate chat data with AI systems using the **Model Context Protocol (MCP)**, enabling contextual interaction with communication data.

Goals

1. Deploy and evaluate a self-hosted Matrix Synapse server.
2. Investigate how MCP can enable AI systems to access chat context.
3. Identify potential automation and knowledge retrieval use cases using chat data.

Specifications

The system will be based on the Matrix ecosystem components:

- **Matrix Protocol** – An open protocol for decentralized messaging and real-time communication.
- **Synapse Server** – The backend implementation of the Matrix protocol responsible for managing users, rooms, and messages.
- **Element Client** – The frontend interface used by users to communicate with the Matrix server.
- **PostgreSQL Database** – Used by Synapse for storing messaging data and metadata.
- **Containerized Deployment** – Synapse and supporting services will be deployed using Docker for simplified setup and scalability.

The project will also explore integration with **MCP**, enabling AI systems to interact with chat data through structured interfaces.

Implementation Plan (8–10 Weeks)

Week 1–2: Matrix Server Setup

- Deploy a **self-hosted Matrix Synapse server** using Docker.
- Configure **PostgreSQL** for storing chat data.
- Connect the **Element client** and create test users.

Week 3: Messaging Validation

- Create chat rooms and test messaging between users.
- Check that messages are stored correctly.
- Explore **Matrix APIs** and understand how chat data is structured.

Week 4–5: Chat Activity Tracking

- Build a small **Python/FastAPI service** to observe chat activity.
- Capture actions such as **new messages, users joining rooms, and reactions**.
- Store this data so it can later be used for **analysis or AI features**.

Week 5–6: AI Context Integration

- Explore **Model Context Protocol (MCP)** to access chat data in a structured way.
- Build a simple AI prototype, such as **chat summarization or question answering based on chat history**.
- Explore possible integrations with other services like **GitHub**.

Week 7–8: Chat Migration

- Investigate whether **existing chats from the current Matrix server** can be migrated.
- If migration is not possible, explore ways to **connect to the existing server or access its data**.